

Name: [REDACTED]Date: [REDACTED]

Chapter 5 Test

Momentum and Collisions

K 9 + 9 18/18	T 18 /18	A 10 /10	C 9 3/4 /10
------------------	-------------	-------------	----------------

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

Excellent work!

THINKING

1. A tennis player hits a serve at a speed of 213 km/h [E], and the opponent returns the 0.056 kg tennis ball to the server with a speed of 127 km/h [W].

- a) Calculate the magnitude of the impulse passed on to the ball by the opponent.
b) Calculate the approximate average force on the ball if the opponent's racquet is in contact with the ball for 0.026 s.

Show all your work.

$$\vec{v}_{i1} = 213 \text{ km/h [E]} = 59.166 \text{ m/s [E]}$$

[4 marks]

$$a) \vec{v}_{i2} = 127 \text{ km/h [W]} = 35.28 \text{ m/s [W]}$$

$$m = 0.056 \text{ kg}$$

$$\Delta v = 35.28 \text{ m/s [W]} = (-59.166 \text{ m/s [E]})$$

$$\Delta p = \text{impulse} = \vec{F} \Delta t = m \Delta v$$

$$\Delta v = 94.446 \text{ m/s [W]}$$

$$\text{impulse} = 0.056 \text{ kg} \cdot 94.446 \text{ m/s [W]}$$

$$\boxed{\text{impulse} = 5.29 \text{ N} \cdot \text{s [W]}}$$

$$b) \vec{F} \Delta t = m \Delta v$$

$$\vec{F} = \frac{m \Delta v}{\Delta t}$$

$$\vec{F} = \frac{0.056 \text{ kg} \cdot 94.446 \text{ m/s [W]}}{0.026 \text{ s}}$$

$$\boxed{\vec{F} = 203.422 \text{ N [W]}}$$

Final answer: a) $5.29 \text{ N} \cdot \text{s [W]}$ Final answer: b) 203.422 N [W]

2. A soccer player in the Qatar 2022 World Cup intends to use his chest to control a ball kicked by another player.

- a) The soccer player first uses his chest to redirect the kicked ball directly upward to a maximum height of 4.0 m above his chest. The mass of the ball is 0.41 kg. Calculate the momentum of the ball when it strikes the player's chest after falling from this height.
- b) The player then uses his chest to bounce the ball straight upward again. If the impact of the ball with the player's chest lasts for 0.2 s, and he imparts an average force of 31 N [up], calculate the speed of the ball after this second collision.

Show all your work.

[5 marks]

a) $h = 4.0 \text{ m [up]}$

$m = 0.41 \text{ kg}$

$E_g = E_k$

$mgh = \frac{1}{2}mv^2$

$v^2 = 2gh$

$v = \sqrt{2gh}$

$v = \sqrt{2 \cdot 9.8 \cdot 4 \text{ m}}$

$v = 8.85 \text{ m/s [down]}$

$\vec{p} = m\vec{v}$

$\vec{p} = 0.41 \text{ kg} \cdot 8.85 \text{ m/s [down]}$

$\vec{p} = 3.63 \text{ kg m/s [down]}$

b) $t = 0.2 \text{ s}$

$F = 31 \text{ N [up]}$

$m = 0.41 \text{ kg}$

$F_{\text{net}} = m\Delta v$

$\Delta v = \frac{F_{\text{net}}}{m}$

$\Delta v = \frac{31 \text{ N} \cdot 0.2 \text{ s}}{0.41 \text{ kg}}$

$\Delta v = 15.12 \text{ m/s [up]}$

$\vec{v}_i = 8.85 \text{ m/s [down]} = 8.85 \text{ m/s [up]}$

$\vec{v}_f = 15.12 \text{ m/s [up]} - 8.85 \text{ m/s [up]}$

$\vec{v}_f = 6.27 \text{ m/s [up]}$

Final answer: a) $3.63 \text{ kg} \cdot \text{m/s [down]}$

Final answer: b) 6.27 m/s [up]

Name: _____

Date: _____

3. Two balls undergo a head-on collision. The ball-1 has a mass of 2.5 kg with an initial velocity of 5 m/s [East] and ball-2 has a mass of 3.5 kg with an initial velocity of 1 m/s [West]. After the head-on collision, the final velocity of ball-1 is 2 m/s [West] and the final velocity of ball-2 is 4 m/s [East]. Using appropriate calculations, what type of collision did the two balls undergo, a perfect elastic collision or perfect inelastic collision? Show all your work.

check if momentum is conserved.

[3 marks]

$$m_1 \vec{v}_{1i} + m_2 \vec{v}_{2i} = m_1 \vec{v}_{1f} + m_2 \vec{v}_{2f}$$

$$m_1 = 2.5 \text{ kg}$$

$$m_2 = 3.5 \text{ kg}$$

$$v_{1i} = 5 \text{ m/s [E]}$$

$$v_{2i} = 1 \text{ m/s [W]} = -1 \text{ m/s [E]}$$

$$v_{1f} = 2 \text{ m/s [W]} = -2 \text{ m/s [E]}$$

$$v_{2f} = 4 \text{ m/s [E]}$$

①

$$2.5 \cdot 5 + 3.5 \cdot (-1) = 2.5 \cdot (-2) + 3.5 \cdot 4$$

$$16 = 9 \text{ kg m/s [E]} = 9 \text{ kg m/s [E]}$$

momentum is conserved.

②

$$\frac{1}{2} m_1 \vec{v}_{1i}^2 + \frac{1}{2} m_2 \vec{v}_{2i}^2 = \frac{1}{2} m_1 \vec{v}_{1f}^2 + \frac{1}{2} m_2 \vec{v}_{2f}^2$$

$$2.5 \cdot (5)^2 + 3.5 \cdot (-1)^2 = 2.5 \cdot (-2)^2 + 3.5 \cdot (4)^2$$

$$66.5$$

$$66.5$$

energy is conserved.

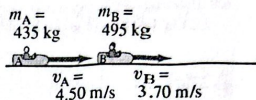
∴ Because both momentum & energy is conserved, the collision must be perfectly elastic.

Final answer: perfect elastic

4. Two bumper cars in an amusement park ride collide elastically as one approaches the other directly from the rear as shown in the diagram. Car A has a mass of 435 kg and car B has a mass of 495 kg, owing to differences in passenger mass. If car A approaches at 4.50 m/s and car B is moving at 3.70 m/s calculate:

- a) the final velocity of Car A and the final velocity of Car B assuming a perfect elastic collision.
b) the change in momentum for Car A and the change in momentum for Car B

Show all your work.



[6 marks]

1. a) $m_1 = 435 \text{ kg}$ ✓
 $m_2 = 495 \text{ kg}$ ✓
 $\vec{v}_{1i} = 4.5 \text{ m/s [F]}$ ✓
 $\vec{v}_{2i} = 3.70 \text{ m/s [F]}$ ✓

$$\vec{v}_{1f} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{v}_{1i} + \left(\frac{2m_2}{m_1 + m_2} \right) \vec{v}_{2i}$$

$$\vec{v}_{1f} = \left(\frac{435 - 495}{435 + 495} \right) 4.5 + \left(\frac{2 \cdot 495}{435 + 495} \right) 3.7$$

$$\therefore \vec{v}_{1f} = 3.65 \text{ m/s [F]}$$

$$\vec{v}_{2f} = \left(\frac{2m_1}{m_1 + m_2} \right) \vec{v}_{1i} + \left(\frac{m_2 - m_1}{m_1 + m_2} \right) \vec{v}_{2i}$$

$$\vec{v}_{2f} = \left(\frac{2 \cdot 435}{435 + 495} \right) 4.5 + \left(\frac{495 - 435}{435 + 495} \right) 3.7$$

$$\therefore \vec{v}_{2f} = 4.45 \text{ m/s [F]}$$

b) $\Delta \vec{p}_1 = m_1 \vec{v}_{1f} - m_1 \vec{v}_{1i}$

$$\Delta \vec{p}_1 = 435 \cdot 3.65 - 435 \cdot 4.5$$

$$\therefore \Delta \vec{p}_1 = -369.75 \text{ kg m/s [F]} \rightarrow 369.75 \text{ kg m/s [B]}$$

$$\Delta \vec{p}_2 = m_2 \vec{v}_{2f} - m_2 \vec{v}_{2i}$$

$$\Delta \vec{p}_2 = 495 \cdot 4.45 - 495 \cdot 3.7$$

$$\therefore \Delta \vec{p}_2 = 371.25 \text{ kg m/s [F]}$$

Final answer: a) 3.65 m/s [F]

Final answer: a) 4.45 m/s [F]

Final answer: b) $\Delta \vec{p}_1 = -369.75 \text{ kg m/s [F]}$

Final answer: b) $\Delta \vec{p}_2 = 371.25 \text{ kg m/s [F]}$

Name: _____

Date: _____

Unit 2 Test

Energy and Momentum

K	T	A	C
/18	/18	10/10	7/10

SHOW ALL YOUR WORK IN ANSWERING QUESTIONS.

APPLICATION

1. The diagram shows a collision between two pucks on an air hockey table. Puck A has a mass of 0.025 kg and is moving along the x axis with a velocity of 5.5 m/s. It makes a collision with puck B, which has a mass of 0.050 kg and is initially at rest. After the collision, the two pucks move apart with the angles shown in the drawing. Determine the final speeds of:

a) puck A

b) puck B

Show all your work.

$$m_A = 0.025 \text{ kg}$$

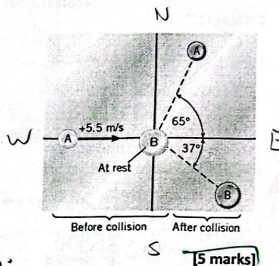
$$m_B = 0.050 \text{ kg}$$

$$v_{Ai} = 5.5 \text{ m/s} [\text{E}]$$

$$v_{Bi} = 0 \text{ m/s}$$

$$\theta_A = 65^\circ$$

$$\theta_B = 37^\circ$$



$$x\text{-momentum: } m_A v_{Ai} = m_A v_{Af} \cos \theta_A + m_B v_{Bf} \cos \theta_B$$

$$0.025 \cdot 5.5 = 0.025 \cdot v_{Af} \cos 65^\circ + 0.05 \cdot v_{Bf} \cos 37^\circ$$

$$0.1375 = 0.010565 v_{Af} + 0.03993 v_{Bf}$$

$$y\text{-momentum: } 0 = m_A v_{Af} \sin \theta_A - m_B v_{Bf} \sin \theta_B$$

$$0 = 0.025 v_{Af} \sin 65^\circ - 0.05 \cdot v_{Bf} \sin 37^\circ$$

$$0 = 0.02266 v_{Af} - 0.03009 v_{Bf}$$

$$v_{Bf} = 1.32806 v_{Af}$$

Sub into x-momentum

$$0.1375 = 0.010565 \cdot 1.32806 v_{Af} + 0.03993 v_{Af}$$

$$v_{Bf} = 2.548 \text{ m/s} [\text{E } 37^\circ \text{ S}]$$

Plug in

$$0 = 0.02266 \cdot v_{Af} - 0.03009 (2.548)$$

$$v_{Af} = 3.384 \text{ m/s} [\text{E } 65^\circ \text{ N}]$$

Final answer: a) $v_{Af} = 3.384 \text{ m/s} [\text{E } 65^\circ \text{ N}]$ Final answer: b) $v_{Bf} = 2.548 \text{ m/s} [\text{E } 37^\circ \text{ S}]$

2. Two balls colliding on a horizontal surface as shown in the diagram, use momentum conservation to determine the magnitude and direction of the final velocity of ball 1.

Show all your work.

$$m_1 = 0.15 \text{ kg}$$

$$v_{1i} = 0.9 \text{ m/s}$$

$$m_2 = 0.26 \text{ kg}$$

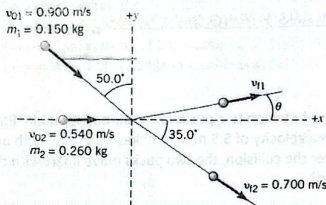
$$v_{2i} = 0.54 \text{ m/s}$$

$$v_{2f} = 0.70 \text{ m/s}$$

$$\theta_{2f} = 35^\circ$$

$$\theta_{1f} = 50^\circ$$

$$\theta_{1f} = 50^\circ$$



This part of the drawing shows the x and y components of the velocity of ball 1 after the collision.

Top view of two balls colliding on a horizontal surface.

[5 marks]

$$v_{1ix} = v_{1i} \cos(\theta_{1i})$$

$$v_{1iy} = -v_{1i} \sin(\theta_{1i})$$

$$v_{2ix} = v_{2i}$$

$$v_{2iy} = 0$$

$$v_{2fx} = v_{2f} \cos(\theta_{2f})$$

$$v_{2fy} = -v_{2f} \sin(\theta_{2f})$$

$$\text{① } x: m_1 v_{1ix} + m_2 v_{2ix} = m_1 v_{1fx} + m_2 v_{2fx}$$

$$v_{1fx} = \frac{(m_1 v_{1ix} + m_2 v_{2ix} - m_2 v_{2fx})}{m_1}$$

plug in

$$v_{1fx} = 0.6313 \text{ m/s}$$

$$\text{② } y: m_1 v_{1iy} + m_2 v_{2iy} = m_1 v_{1fy} + m_2 v_{2fy}$$

$$v_{1fy} = \frac{(m_1 v_{1iy} + m_2 v_{2iy} - m_2 v_{2fy})}{m_1}$$

plug in

$$v_{1fy} = 0.113 \text{ m/s}$$

$$\tan \theta = \frac{v_{1fy}}{v_{1fx}}$$

$$\theta = \tan^{-1} \left(\frac{0.113}{0.6313} \right)$$

$$\therefore \theta = 10.59^\circ$$

$$v_{1f}^2 = v_{1fx}^2 + v_{1fy}^2$$

$$v_{1f}^2 = 0.41246$$

$$v_{1f} = 0.642 \text{ m/s}$$

$$\text{Final answer: } \vec{v}_1 = 0.642 \text{ m/s } [E 10.6^\circ N]$$

COMMUNICATION

1. Can a centripetal force ever do work on an object if the object is moving in a circle? Explain.

No, work is calculated with the equation $W = F \cdot d \cos \theta$. (2 marks)
 The centripetal force always points towards the center of a object travelling in a circle, thus making it perpendicular to its instantaneous displacement. This means the θ is 90° , which when plugged into $\cos \theta$, $= 0$, which means no work is done by centripetal forces.

2. For each situation, state whether the collision is elastic or inelastic with explanation:

a) You drop a ball from your hand. It collides with the floor and bounces back up so that it just reaches your hand

b) You drop a ball of clay from your hand. When it collides with the ground, it stops.

a) the ball's collision with the ground is elastic because it maintains its momentum and kinetic energy, returning to the same height as before. (2 marks)
 b) the clay's momentum stays the same upon collision, but it loses kinetic energy to deformation, which means the collision is inelastic.

3. A car has the same kinetic energy when it is traveling south at 30 m/s as when it is traveling north at 30 m/s. Is the momentum of the car the same in both cases? Explain.

No, because momentum has a direction and is a vector, while kinetic energy is scalar. When an object with mass 500 kg is traveling at 30 m/s , the kinetic energy would be the same regardless of direction, being 22500 J , but the momentum would be $15000 \text{ kg}\cdot\text{m/s}$ [S] and $15000 \text{ kg}\cdot\text{m/s}$ [N]. (2 marks)

4. A squash ball hits a wall at a 45° angle as shown in the diagram. Determine the direction:

i) of the change in momentum of the ball. Explain and show all your work.

$$V_f = V_i$$

ii) of the force on the wall. Explain and show all your work.

assume elastic collision

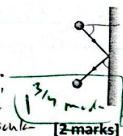
i) $\Delta p_x = m(-v \cos 45^\circ - v \cos 45^\circ) = -2mv \cos 45^\circ$

$\Delta p_y = m(-v \sin 45^\circ - v \sin 45^\circ) = 0$

$\tan \theta = \frac{\Delta p_y}{\Delta p_x} = 0$; only horizontal change

is perpendicular to the wall

Communication marks are awarded throughout the test according to the shown rubric. (2 marks)



	Level 1	Level 2	Level 3	Level 4
Correct use of mathematical/ physical symbols, labels, and conventions (e.g. units, proper component form for the values, positive direction given, etc...)	Few mathematical/ physical conventions are used correctly.	Some mathematical/ physical conventions are used correctly.	Most mathematical/ physical conventions are used correctly.	Consistently uses mathematical/ physical conventions correctly.
	[1/2]	[1]	[1-1/2]	[2]